

Week 4 — Probability Review, and the Markowitz Portfolio

AD7031 Optimization Under Uncertainty • Instructor: Prof. H. Park • Sep 19, 2026

1. Random variables

back to week 2's Section 2 table: the OR column had **uncertain data** ξ . today, the vocabulary — written in this course's notation from the first line — x is already taken by the **decision**, so the uncertain data is $\tilde{\xi}$ throughout.

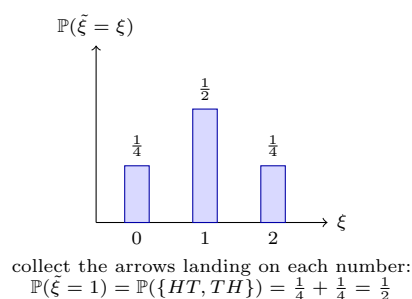
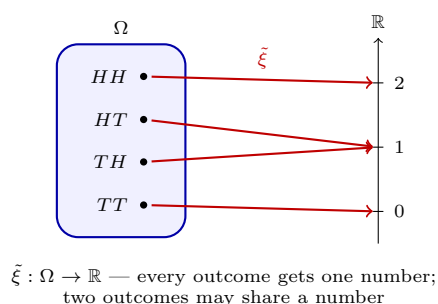
- **random experiment**: its outcome is not known in advance
- **sample space** Ω = the set of all possible outcomes

Def (*random variable*) a function that assigns a real number to each outcome: $\tilde{\xi} : \Omega \rightarrow \mathbb{R}$

the tilde marks the *random* object; ξ without it is a **realization** — one value it took

toss a coin twice: $\Omega = \{HH, HT, TH, TT\}$, each outcome has probability $\frac{1}{4}$

$\tilde{\xi}$ = number of heads: $\tilde{\xi}(HH) = 2$, $\tilde{\xi}(HT) = 1$, $\tilde{\xi}(TH) = 1$, $\tilde{\xi}(TT) = 0$



- **discrete** $\tilde{\xi}$: countably many values — number of defective parts, number of arrivals, demand
- **continuous** $\tilde{\xi}$: values fill an interval — length, time, temperature, a return
- **support** Ξ = the set of values $\tilde{\xi}$ actually takes (the smallest closed set with $\mathbb{P}(\tilde{\xi} \in \Xi) = 1$)
— Ω is where the *experiment* lives, $\Xi \subseteq \mathbb{R}^k$ is where the *numbers* live

2. Events, conditioning, independence

an **event** is a subset of the sample space, $A \subseteq \Omega$; $\mathbb{P}(A)$ is the probability it occurs.

- $0 \leq \mathbb{P}(A) \leq 1$, $\mathbb{P}(\Omega) = 1$
- $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$ — subtract the overlap, or it is counted twice

Def (*conditional probability*) $\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$, defined when $\mathbb{P}(B) > 0$

— B becomes the new sample space; divide by $\mathbb{P}(B)$ so the weights add to 1 again

Def (*independent events*) A and B are independent if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$

then $\mathbb{P}(A | B) = \frac{\mathbb{P}(A)\mathbb{P}(B)}{\mathbb{P}(B)} = \mathbb{P}(A)$ — learning B tells you nothing about A

Thm (*law of total probability*) let $A_1, \dots, A_I$ **partition** Ω : $A_i \cap A_j = \emptyset$ for $i \neq j$, and $\bigcup_i A_i = \Omega$. then for any event B ,

$$\mathbb{P}(B) = \mathbb{P}(B \cap A_1) + \dots + \mathbb{P}(B \cap A_I) = \mathbb{P}(B | A_1)\mathbb{P}(A_1) + \dots + \mathbb{P}(B | A_I)\mathbb{P}(A_I)$$

— split the world into cases, answer each case, weight by how likely the case is

two coins again. $A_1 = \text{first toss is } H$, $A_2 = \text{first toss is } T$ — a partition, each of probability $\frac{1}{2}$

take $B = \{\tilde{\xi} \geq 1\}$: $\mathbb{P}(B | A_1) = 1$ (a head already landed), $\mathbb{P}(B | A_2) = \frac{1}{2}$ (needs the second)

$\mathbb{P}(B) = 1 \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4}$ — the same $\frac{3}{4}$ the PMF gives in Section 3

takeaway conditioning is how a *two-stage* decision gets written down: act, watch which A_i occurred, act again. weeks 10–12 run on this.

3. Discrete random variables

a discrete $\tilde{\xi}$ is a **list**: finitely many values, each carrying a weight.

Def (*scenarios*) $\tilde{\xi}$ takes the values $\xi^1, \dots, \xi^S$ — the **scenarios** — with

$$p_s = \mathbb{P}(\tilde{\xi} = \xi^s), \quad (1) \ p_s \geq 0 \quad (2) \ \sum_{s \in [S]} p_s = 1, \quad [S] = \{1, \dots, S\}$$

$$\mathbb{P}(\tilde{\xi} \in A) = \sum_{s: \xi^s \in A} p_s \quad \text{— add the weights of the scenarios landing in } A$$

the scenario index goes up: ξ^s is scenario s ; a subscript ξ_j is component j of a vector (Section 8). two different jobs, two different slots.

older books call this list the **probability mass function** and write it as a function f . same content — but a table, and a table is the only thing a solver can be handed.

$$\text{two coins: } S = 3, \quad \xi^1 = 0, \ \xi^2 = 1, \ \xi^3 = 2, \quad p = \left(\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right)$$

$$\mathbb{P}(\tilde{\xi} \geq 1) = p_2 + p_3 = \frac{3}{4} \quad \text{— the } \frac{3}{4} \text{ Section 2 reached by conditioning}$$

Def (*mean, variance*)

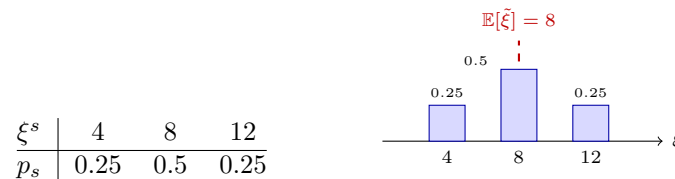
$$\text{mean: } \mu = \mathbb{E}[\tilde{\xi}] = \sum_{s \in [S]} p_s \xi^s$$

$$\text{variance: } \sigma^2 = \text{Var}(\tilde{\xi}) = \sum_{s \in [S]} p_s (\xi^s - \mu)^2 = \mathbb{E}[\tilde{\xi}^2] - \mu^2, \quad \sigma = \sqrt{\text{Var}(\tilde{\xi})}$$

$$\text{function of } \tilde{\xi}: \quad \mathbb{E}[h(\tilde{\xi})] = \sum_{s \in [S]} p_s h(\xi^s) \quad \leftarrow \text{law of the unconscious statistician}$$

$$\mathbb{E}[a\tilde{\xi} + b] = a \mathbb{E}[\tilde{\xi}] + b, \quad \text{Var}(a\tilde{\xi} + b) = a^2 \text{Var}(\tilde{\xi})$$

Example 1 demand $\tilde{\xi}$ for a spare part next month:



$$\mathbb{E}[\tilde{\xi}] = 0.25 \cdot 4 + 0.5 \cdot 8 + 0.25 \cdot 12 = 8$$

$$\text{Var}(\tilde{\xi}) = 0.25 \cdot (4 - 8)^2 + 0.5 \cdot (8 - 8)^2 + 0.25 \cdot (12 - 8)^2 = 8, \quad \sigma = 2\sqrt{2} \approx 2.83$$

4. Continuous random variables

Def (*probability density function, PDF*) $f : \mathbb{R} \rightarrow \mathbb{R}_+$ with

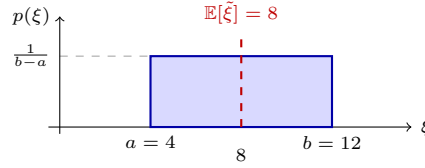
$$(1) \ p(\xi) \geq 0 \quad (2) \ \int_{-\infty}^{\infty} p(\xi) d\xi = 1 \quad (3) \ \mathbb{P}(a \leq \tilde{\xi} \leq b) = \int_a^b p(\xi) d\xi = \text{area under } p \text{ from } a \text{ to } b$$

$p(\xi)$ is *not* a probability: $\mathbb{P}(\tilde{\xi} = \xi) = 0$ for every single ξ , and $p(\xi) > 1$ is allowed

Def (mean, variance) same as discrete — the sums become integrals:

$$\mu = \mathbb{E}[\tilde{\xi}] = \int \xi p(\xi) d\xi, \quad \sigma^2 = \text{Var}(\tilde{\xi}) = \int (\xi - \mu)^2 p(\xi) d\xi = \mathbb{E}[\tilde{\xi}^2] - \mu^2, \quad \mathbb{E}[h(\tilde{\xi})] = \int h(\xi) p(\xi) d\xi$$

Example 2 $\tilde{\xi} \sim \text{Uniform}[a, b]$ — $p(\xi) = \frac{1}{b-a}$ on $[a, b]$, 0 outside



mean:

$$\mathbb{E}[\tilde{\xi}] = \int_a^b \xi \frac{1}{b-a} d\xi = \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b = \frac{b^2 - a^2}{2(b-a)} = \frac{a+b}{2}$$

second moment — $h(x) = x^2$:

$$\mathbb{E}[\tilde{\xi}^2] = \int_a^b \xi^2 \frac{1}{b-a} d\xi = \frac{1}{b-a} \left[\frac{x^3}{3} \right]_a^b = \frac{b^3 - a^3}{3(b-a)} = \frac{a^2 + ab + b^2}{3}$$

variance — subtract, no new integral:

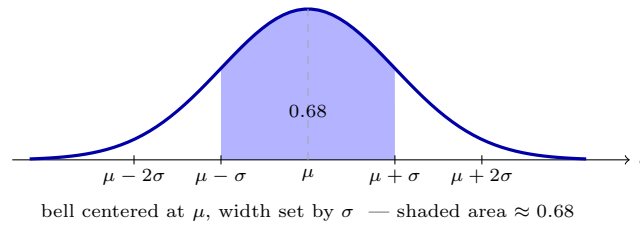
$$\text{Var}(\tilde{\xi}) = \mathbb{E}[\tilde{\xi}^2] - \mathbb{E}[\tilde{\xi}]^2 = \frac{a^2 + ab + b^2}{3} - \frac{(a+b)^2}{4} = \frac{(b-a)^2}{12}$$

$$\text{with } a=4, b=12: \quad \mathbb{E}[\tilde{\xi}] = 8, \quad \text{Var}(\tilde{\xi}) = \frac{64}{12} = \frac{16}{3}, \quad \sigma = \frac{4}{\sqrt{3}} \approx 2.31$$

Normal distribution $\tilde{\xi} \sim \mathcal{N}(\mu, \sigma^2)$: $p(\xi) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(\xi - \mu)^2}{2\sigma^2}\right)$, $\xi \in \mathbb{R}$

$$\mathbb{E}[\tilde{\xi}] = \mu, \quad \text{Var}(\tilde{\xi}) = \sigma^2$$

$$\mathbb{P}(\mu - \sigma \leq \tilde{\xi} \leq \mu + \sigma) \approx 0.68, \quad \mathbb{P}(\mu - 2\sigma \leq \tilde{\xi} \leq \mu + 2\sigma) \approx 0.95$$



5. Expectation, in one formula

Sections 3 and 4 defined the mean twice — once a sum, once an integral. one definition covers both.

Def (expectation) for $\tilde{\xi} \sim \mathbb{P}$,

$$\mathbb{E}_{\mathbb{P}}[\tilde{\xi}] = \int_{\Omega} \xi \mathbb{P}(d\xi)$$

read $\mathbb{P}(d\xi)$ as *the probability sitting on a small piece $d\xi$* — the weight, whichever form it takes:

	the weight $\mathbb{P}(d\xi)$ is	and $\mathbb{E}_{\mathbb{P}}[\tilde{\xi}]$ reads
discrete	p_s , one per scenario ξ^s	$\sum_{s \in [S]} p_s \xi^s$
continuous	$p(\xi) d\xi$	$\int_{\Xi} \xi p(\xi) d\xi$

Def (*expectation of a function*) for $f : \mathbb{R}^k \rightarrow \mathbb{R}$ and a random vector $\tilde{\xi}$,

$$\mathbb{E}_{\mathbb{P}}[f(\tilde{\xi})] = \int_{\Omega} f(\xi) \mathbb{P}(d\xi) \quad \text{---} \quad \sum_{s \in [S]} p_s f(\xi^s) \quad \text{or} \quad \int_{\Xi} f(\xi) p(\xi) d\xi$$

this is the one the course actually needs: f will be a **cost**, and $\mathbb{E}[f(x, \tilde{\xi})]$ the objective.

variance is an expectation too, so it inherits the same two readings:

$$\text{Var}(\tilde{\xi}) = \mathbb{E}[(\tilde{\xi} - \mathbb{E}[\tilde{\xi}])^2], \quad \sigma(\tilde{\xi}) = \sqrt{\text{Var}(\tilde{\xi})}, \quad \text{Cov}(\tilde{\xi}_1, \tilde{\xi}_2) = \mathbb{E}[(\tilde{\xi}_1 - \mathbb{E}[\tilde{\xi}_1])(\tilde{\xi}_2 - \mathbb{E}[\tilde{\xi}_2])]$$

takeaway one definition, two computations. write $\mathbb{E}$ once and stop asking whether the data is discrete or continuous.

6. Two random variables

Def (*joint PMF*) $p(\xi_1, \xi_2) = \mathbb{P}(\tilde{\xi}_1 = \xi_1, \tilde{\xi}_2 = \xi_2), \quad p \geq 0, \quad \sum_x \sum_y p(\xi_1, \xi_2) = 1$

Def (*marginal PMF*) $p_1(\xi_1) = \sum_y p(\xi_1, \xi_2), \quad p_2(\xi_2) = \sum_x p(\xi_1, \xi_2)$

— row sums and column sums, written in the *margins* of the table

Example 3 two assets, returns tomorrow: $\tilde{\xi}_1, \tilde{\xi}_2$ each either down (-0.10) or up ($+0.20$)

$p(\xi_1, \xi_2)$	$\xi_2 = -0.10$	$\xi_2 = 0.20$	$p_1(\xi_1)$
$\xi_1 = -0.10$	0.3	0.2	0.5
$\xi_1 = 0.20$	0.2	0.3	0.5
$p_2(\xi_2)$	0.5	0.5	1

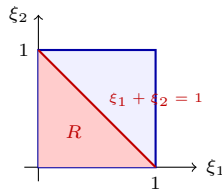
each asset alone: down or up w.p. $\frac{1}{2}$. together: same direction w.p. $0.3 + 0.3 = 0.6$

Def (*joint PDF*) $p : \mathbb{R}^2 \rightarrow \mathbb{R}_+, \quad \iint p(\xi_1, \xi_2) d\xi_1 d\xi_2 = 1, \quad \mathbb{P}((\tilde{\xi}_1, \tilde{\xi}_2) \in R) = \iint_R p(\xi_1, \xi_2) d\xi_1 d\xi_2$

— the **volume** under p above the region R

marginal PDF: $p_1(\xi_1) = \int p(\xi_1, \xi_2) dy$ — the sum becomes an integral

Example 4 $(\tilde{\xi}_1, \tilde{\xi}_2)$ uniform on the unit square: $p(\xi_1, \xi_2) = 1$ on $[0, 1]^2$, 0 outside



$$\mathbb{P}(\tilde{\xi}_1 + \tilde{\xi}_2 \leq 1) = \iint_R 1 d\xi_1 d\xi_2 = \text{area of the triangle } R = \frac{1}{2}$$

$$\text{marginal: } p_1(\xi_1) = \int_0^1 1 d\xi_2 = 1 \text{ on } [0, 1] \text{ — } \tilde{\xi}_1 \text{ alone is Uniform}[0, 1]$$

Def (*independence*) $\tilde{\xi}_1, \tilde{\xi}_2$ are independent if $p(\xi_1, \xi_2) = p_1(\xi_1) p_2(\xi_2) \quad \forall \xi_1, \xi_2$ (PMF or PDF)

Example 3: $p(-0.10, -0.10) = 0.3 \neq 0.5 \cdot 0.5$ — **not** independent

Example 4: $p(\xi_1, \xi_2) = 1 = 1 \cdot 1$ — independent

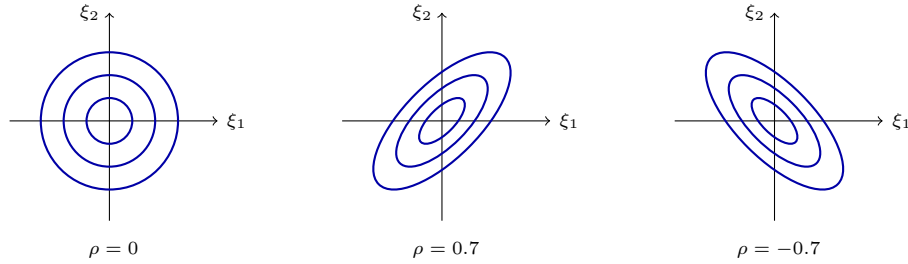
7. Covariance and correlation

Def (*covariance, correlation*)

$$\text{Cov}(\tilde{\xi}_1, \tilde{\xi}_2) = \mathbb{E}[(\tilde{\xi}_1 - \mu_1)(\tilde{\xi}_2 - \mu_2)] = \mathbb{E}[\tilde{\xi}_1 \tilde{\xi}_2] - \mu_1 \mu_2, \quad \text{Cov}(\tilde{\xi}_1, \tilde{\xi}_1) = \text{Var}(\tilde{\xi}_1)$$

$$\rho_{12} = \frac{\text{Cov}(\tilde{\xi}_1, \tilde{\xi}_2)}{\sigma_1 \sigma_2}, \quad -1 \leq \rho_{12} \leq 1$$

$\rho > 0$: move together, $\rho < 0$: move opposite, independent $\Rightarrow \rho = 0$



contours of the joint PDF of two normal random variables, $\mu_1 = \mu_2 = 0$, $\sigma_1 = \sigma_2 = 1$

Example 3, continued (use the margins for μ and σ , all four cells for the covariance)

$$\mu_1 = 0.5 \cdot (-0.10) + 0.5 \cdot 0.20 = 0.05, \quad \mu_2 = 0.05$$

$$\text{Var}(\tilde{\xi}_1) = 0.5 \cdot (-0.15)^2 + 0.5 \cdot (0.15)^2 = 0.0225, \quad \sigma_1 = \sigma_2 = 0.15$$

$$\begin{aligned} \text{Cov}(\tilde{\xi}_1, \tilde{\xi}_2) &= 0.3(-0.15)(-0.15) + 0.2(-0.15)(0.15) + 0.2(0.15)(-0.15) + 0.3(0.15)(0.15) \\ &= (0.3 - 0.2 - 0.2 + 0.3) \cdot 0.0225 = 0.0045 \end{aligned}$$

$$\rho_{12} = \frac{0.0045}{0.15 \cdot 0.15} = 0.2 \quad \text{— the two assets tend to move together}$$

8. Linear combinations

$$\mathbb{E}[a\tilde{\xi}_1 + b\tilde{\xi}_2] = a\mu_1 + b\mu_2 \quad \text{— always, no independence needed}$$

$$\text{Var}(a\tilde{\xi}_1 + b\tilde{\xi}_2) = a^2 \text{Var}(\tilde{\xi}_1) + b^2 \text{Var}(\tilde{\xi}_2) + 2ab \text{Cov}(\tilde{\xi}_1, \tilde{\xi}_2)$$

n random variables $\tilde{\xi} = (\tilde{\xi}_1, \dots, \tilde{\xi}_n)$:

$$\text{mean vector } \mu = (\mathbb{E}[\tilde{\xi}_1], \dots, \mathbb{E}[\tilde{\xi}_n])^\top \in \mathbb{R}^n$$

covariance matrix $\Sigma \in \mathbb{R}^{n \times n}$, $\Sigma_{ij} = \text{Cov}(\tilde{\xi}_i, \tilde{\xi}_j)$; diagonal $\Sigma_{ii} = \text{Var}(\tilde{\xi}_i)$; symmetric; always $\Sigma \succeq 0$

$$\text{for fixed } c \in \mathbb{R}^n: \quad \mathbb{E}[c^\top \tilde{\xi}] = c^\top \mu, \quad \text{Var}(c^\top \tilde{\xi}) = c^\top \Sigma c$$

$$n = 2, \quad c = (a, b): \quad c^\top \Sigma c = \begin{pmatrix} a & b \end{pmatrix} \begin{pmatrix} \text{Var}(\tilde{\xi}_1) & \text{Cov}(\tilde{\xi}_1, \tilde{\xi}_2) \\ \text{Cov}(\tilde{\xi}_1, \tilde{\xi}_2) & \text{Var}(\tilde{\xi}_2) \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} \quad \text{— Var}(a\tilde{\xi}_1 + b\tilde{\xi}_2) \text{ again}$$

$$\text{Example 3: } \mu = \begin{pmatrix} 0.05 \\ 0.05 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 0.0225 & 0.0045 \\ 0.0045 & 0.0225 \end{pmatrix}$$

9. The bridge to optimization

everything so far was about $\tilde{\xi}$ alone. now put the decision back in.

a cost depends on both: $f(x, \tilde{\xi})$ — x the **decision** we choose, $\tilde{\xi}$ the **data** we do not

fix x . then $f(x, \tilde{\xi})$ is a **random variable** — it inherits $\tilde{\xi}$'s randomness

so $\mathbb{E}[f(x, \tilde{\xi})]$ is a **number**, one per x — and numbers can be minimized:

$$\min_{x \in X} \mathbb{E}[f(x, \tilde{\xi})]$$

that $\mathbb{E}$ is the whole design space of this course. expectation is one way to turn a random cost into a number; CVaR, VaR and a chance constraint are others. the model above changes in that one slot and nowhere else.

takeaway every paradigm in this course differs in one place only: *how it collapses the randomness to a number.*

10. Running example: the Markowitz portfolio

a budget of 1, n assets. decide the split today; the returns are revealed tomorrow.

- **decision:** $x \in \mathbb{R}^n$, x_j = share of the budget in asset j
- **uncertain data:** $\tilde{\xi} = (\tilde{\xi}_1, \dots, \tilde{\xi}_n) \in \mathbb{R}^n$, $\tilde{\xi}_j$ = return of asset j — a random vector with mean μ , covariance matrix Σ
- **feasible set:** $X = \{x \in \mathbb{R}^n : \mathbf{1}^\top x = 1, x \geq 0\}$ ($\mathbf{1}$ = all-ones vector, $\mathbf{1}^\top x = x_1 + \dots + x_n$)

Portfolio return $\tilde{\xi}^\top x = x_1 \tilde{\xi}_1 + \dots + x_n \tilde{\xi}_n$ — a random variable

as a function: fix x , $f(\xi) = \xi^\top x$, $f : \mathbb{R}^n \rightarrow \mathbb{R}$ — the week 2 note's inner product, fed a random input

loss: $f(x, \tilde{\xi}) = -\tilde{\xi}^\top x$ (a gain is a negative loss)

Section 8's linear combinations with $c = x$: $\mathbb{E}[\tilde{\xi}^\top x] = \mu^\top x$, $\text{Var}(\tilde{\xi}^\top x) = x^\top \Sigma x$

Check on Example 3 $(\tilde{\xi}_1, \tilde{\xi}_2) = \text{Example 3's } (\tilde{\xi}_1, \tilde{\xi}_2)$, half in each: $x = (\frac{1}{2}, \frac{1}{2})$

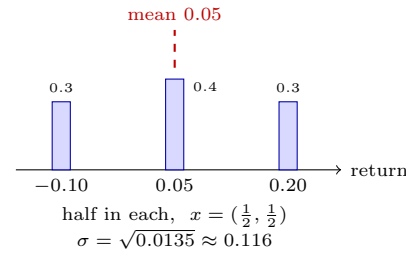
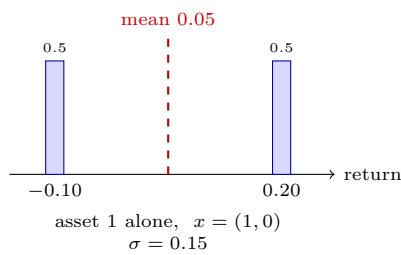
ξ_1	ξ_2	$\frac{1}{2}\xi_1 + \frac{1}{2}\xi_2$	probability	$\Rightarrow$ PMF of $\tilde{\xi}^\top x$:	value		
-0.10	-0.10	-0.10	0.3			-0.10	0.05
-0.10	0.20	0.05	0.2		probability	0.3	0.4
0.20	-0.10	0.05	0.2			0.20	0.3
0.20	0.20	0.20	0.3				

from the PMF: $\mathbb{E}[\tilde{\xi}^\top x] = 0.3(-0.10) + 0.4(0.05) + 0.3(0.20) = 0.05$

$\text{Var}(\tilde{\xi}^\top x) = 0.3(-0.15)^2 + 0.4(0)^2 + 0.3(0.15)^2 = 0.0135$

from μ and Σ : $\mu^\top x = \frac{1}{2}(0.05) + \frac{1}{2}(0.05) = 0.05 \checkmark$

$x^\top \Sigma x = \frac{1}{4}(0.0225) + \frac{1}{4}(0.0225) + 2 \cdot \frac{1}{4}(0.0045) = 0.0135 \checkmark$



same mean, less spread: **diversification**. it comes from $\rho < 1$ — with $\rho = 1$, $x^\top \Sigma x$ would stay 0.0225.

The model (Markowitz, 1952) — least risk, subject to an expected-return target r :

$$\begin{aligned}
 \min_x \quad & x^\top \Sigma x && \text{variance of the return (risk)} \\
 \text{s.t.} \quad & \mu^\top x \geq r && \text{expected return meets the target} \\
 & \mathbf{1}^\top x = 1, \quad x \geq 0 && x \in X
 \end{aligned}$$

Section 5's QP with $Q = 2\Sigma$, $c = 0$; convex because $\Sigma \succeq 0$ (Section 6).

in the language of the bridge: the loss $-\tilde{\xi}^\top x$ is random; Markowitz collapses it with two numbers — its variance (objective) and its mean (constraint).

Data — the coding session $n = 3$, $r = 0.10$

asset j	expected return μ_j	variance Σ_{jj}	$\sigma_j = \sqrt{\Sigma_{jj}}$
bond	0.06	0.010	0.10
tech	0.10	0.040	0.20
energy	0.15	0.090	0.30

$$x = \begin{pmatrix} x_{\text{bond}} \\ x_{\text{tech}} \\ x_{\text{energy}} \end{pmatrix}, \quad \mu = \begin{pmatrix} 0.06 \\ 0.10 \\ 0.15 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 0.010 & 0.001 & 0 \\ 0.001 & 0.040 & 0.015 \\ 0 & 0.015 & 0.090 \end{pmatrix}$$

covariances: bond–tech 0.001, tech–energy 0.015, bond–energy 0

written out term by term — exactly the lines typed in the code:

$$\begin{aligned} \min \quad & 0.010 x_{\text{bond}}^2 + 0.040 x_{\text{tech}}^2 + 0.090 x_{\text{energy}}^2 + 2(0.001) x_{\text{bond}} x_{\text{tech}} + 2(0.015) x_{\text{tech}} x_{\text{energy}} \\ \text{s.t.} \quad & 0.06 x_{\text{bond}} + 0.10 x_{\text{tech}} + 0.15 x_{\text{energy}} \geq 0.10 \\ & x_{\text{bond}} + x_{\text{tech}} + x_{\text{energy}} \leq 1 \\ & x_{\text{bond}}, x_{\text{tech}}, x_{\text{energy}} \geq 0 \end{aligned}$$

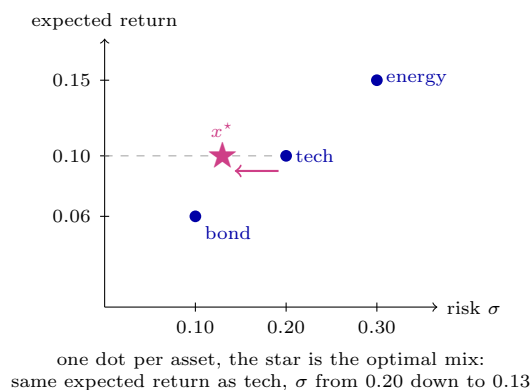
budget “ ≤ 1 ”: leftover sits in cash earning 0 — never helps, so the sum is 1 at the optimum

Solution

$$x^* \approx (0.39, 0.29, 0.31), \quad x^{*\top} \Sigma x^* \approx 0.0169, \quad \sigma \approx 0.130 \quad \mu^\top x^* = 0.10 \text{ — target met exactly}$$

all in tech: same expected return 0.10, but $\sigma = 0.20$. the mix cuts σ by about a third.

every asset gets a share — not a vertex (Section 5).



demo `week03_markowitz.ipynb` — Colab — build this model live from the skeleton, then compare with the completed cell.

demo `demos/markowitz_portfolio.py` — the same model as a plain script: prints the model, solves it, plots the weights x^* .

takeaway the running example: the loss $-\tilde{\xi}^\top x$ stays; each later model changes only how it is collapsed to a number — today variance, next CVaR and VaR.

next no class Sep 26. week 6 — duality: every LP comes with a shadow LP, and why $\max = \min$.